

Diversity Healthcare

Phase 2

Contributed by:

Anabelle Hsiao

Zeynep Sude Haksal

Christian Rogers

Muhammad Bilal

Vignesh Veerakumar Sathya

Ho Ching Ashley Wong

1. Persona(s), Scenario(s) and HTA based on the problem or solution conceptualized in Phase I.

Personas:

Patient Persona – Sarah (32):



SARAH (32)

CHARACTERISTICS:

HARD WORKING, PASSIONATE, EFFICIENT.

INTERESTS:

WATCHING MARKETING STRATEGY VIDEOS ON YOUTUBE, YOGA, AND LISTENING TO AUDIO BOOKS

CAREER:

MARKETING MANAGER FROM TORONTO

ROLE ON CLINIC APP:

PATIENT

FRUSTRATION:

SARAH IS A BUSY PERSON WHO STRUGGLES TO BALANCE WORK, FAMILY, AND HEALTHCARE. HER MAIN FRUSTRATION IS REPEATING MEDICAL INFORMATION AND WAITING FOR RESULTS.

NEED/MOTIVATION:

SHE WANTS AN EASY WAY TO BOOK APPOINTMENTS, VIEW TEST RESULTS, AND MANAGE PRESCRIPTIONS IN ONE PLACE. HENCE, CONVENIENCE AND ACCESS TO HER HEALTH RECORDS ARE VALUABLE TO HER. SHE IS MOTIVATED BY IMPROVING EFFICIENCY AND ACCESSIBILITY TO HER HEALTH LIFE THROUGH THIS APP

Doctor Persona – Dr. Mike (45):



DR. MIKE (45)

CHARACTERISTICS:

PATIENT, CARING, AND VALUES EFFICIENCY

INTERESTS:

BINGE WATCHING CRIME DOCUMENTARIES ON FRIDAY NIGHTS, DOG WALKS, ARCHERY, AND SINGING

CAREER:

A FAMILY PHYSICIAN IN MISSISSAUGA WHO SEES OVER 20 PATIENTS A DAY

ROLE ON CLINIC APP:

DOCTOR

FRUSTRATION:

HIS MAIN FRUSTRATION IS DEALING WITH SLOW, COMPLEX EMR SYSTEMS.

NEED/MOTIVATION:

HE'S MOTIVATED BY IMPROVING PATIENT CARE THROUGH BETTER TECHNOLOGY. HE WANTS A SYSTEM THAT SIMPLIFIES PATIENT RECORD MANAGEMENT, SUPPORTS TELEMEDICINE, AND REDUCES ADMINISTRATIVE TIME.

Scenarios:

Tasks

1. view/ book/ cancel appointments
2. Dr view patient records
3. Prescription records saved
4. View clinics around the area
5. Change user information (account) such as SIN number and log in password

The Story of Sarah:

On Oct 20th, Toronto-based marketer Sarah(32 years old) is packing for a same-day flight to Edmonton, where her team will host a celebration for closing the Nike deal—a milestone she led and must attend. While checking in online, she notices her Toronto dental cleaning at [8 p.m. ET] will overlap once she lands and begins event prep ([5:30 p.m. MT] flight arrival). From her phone, she opens the clinic portal, authenticates with SIN and password, and uses the calendar view to confirm the time-zone conflict. The system flags the clash, shows the cancellation policy (fee waived if rebooked), and offers a “Rebook in Destination City” option using her itinerary’s arrival airport. She filters by Location = Edmonton, Date = [tomorrow], Time after 4:00 p.m. MT, and finds an opening near her hotel at [5:15 p.m. MT]. With one tap, she cancels the Toronto slot, books the Edmonton clinic, and opts into SMS/email confirmations; her calendar updates automatically, travel buffer is added, and her benefits tracker notes the provider change. The system logs a clean audit trail—turning a cross-city, cross-time-zone conflict into a seamless reschedule aligned with her trip.

[condition 1,4,5-2 are satisfied]

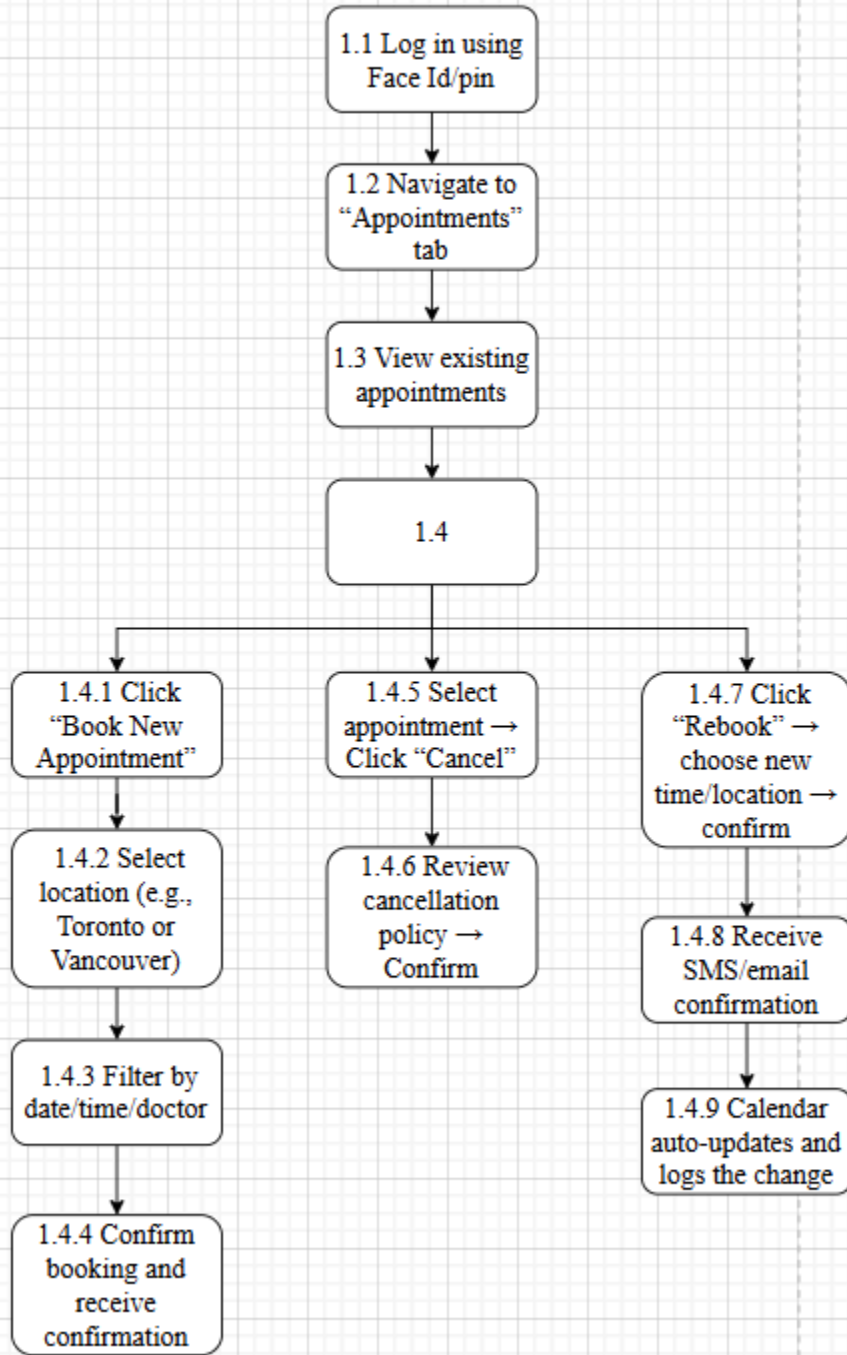
The Story of Dr.Mike:

Dr. Mike, a family physician in Mississauga who sees 20+ patients daily. Today, on Nov 17th, he opens the clinic portal for a telemedicine follow-up, e-prescribes an inhaler, and the system auto-saves the prescription to the chart with a timestamped, versioned record. He then opens the calendar view to verify all appointments for the week. During a quick admin pause, he updates his contact email in Settings, verifies it via a one-click link, and all notifications (labs, e-Rx ACKs, appointment changes) switch to the new address. By lunch, he’s issued clean prescription records, verified appointments without phone tag, and fixed his email—all in one smooth flow that trades the usual slow, complex EMR pain for real efficiency.

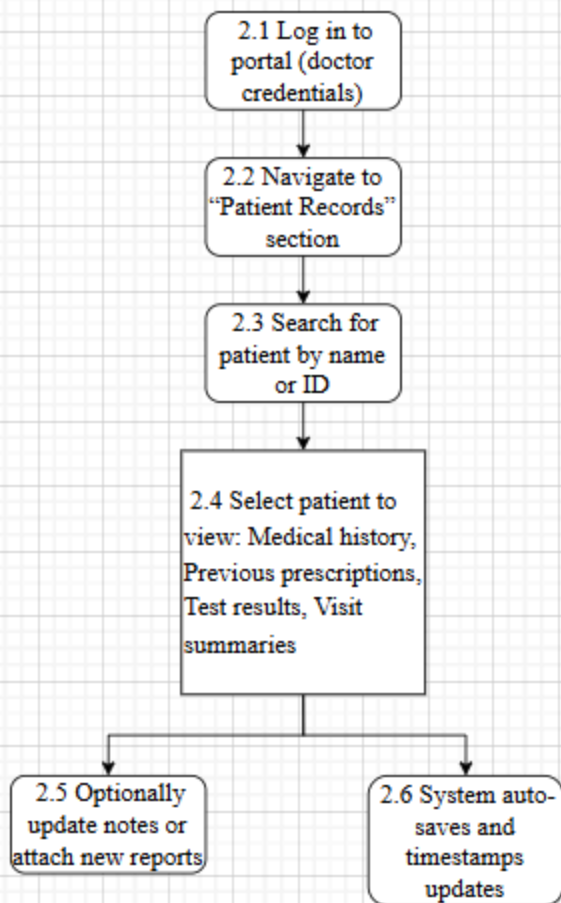
[condition 2,3,5-1 are satisfied]

HTA

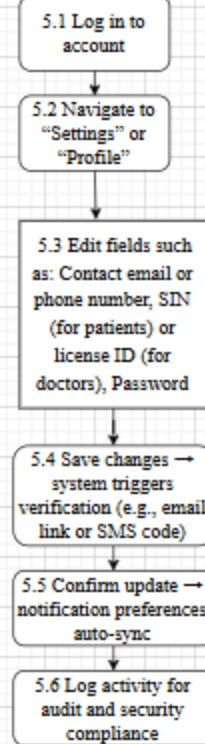
View/Book
Cancel
Appointments

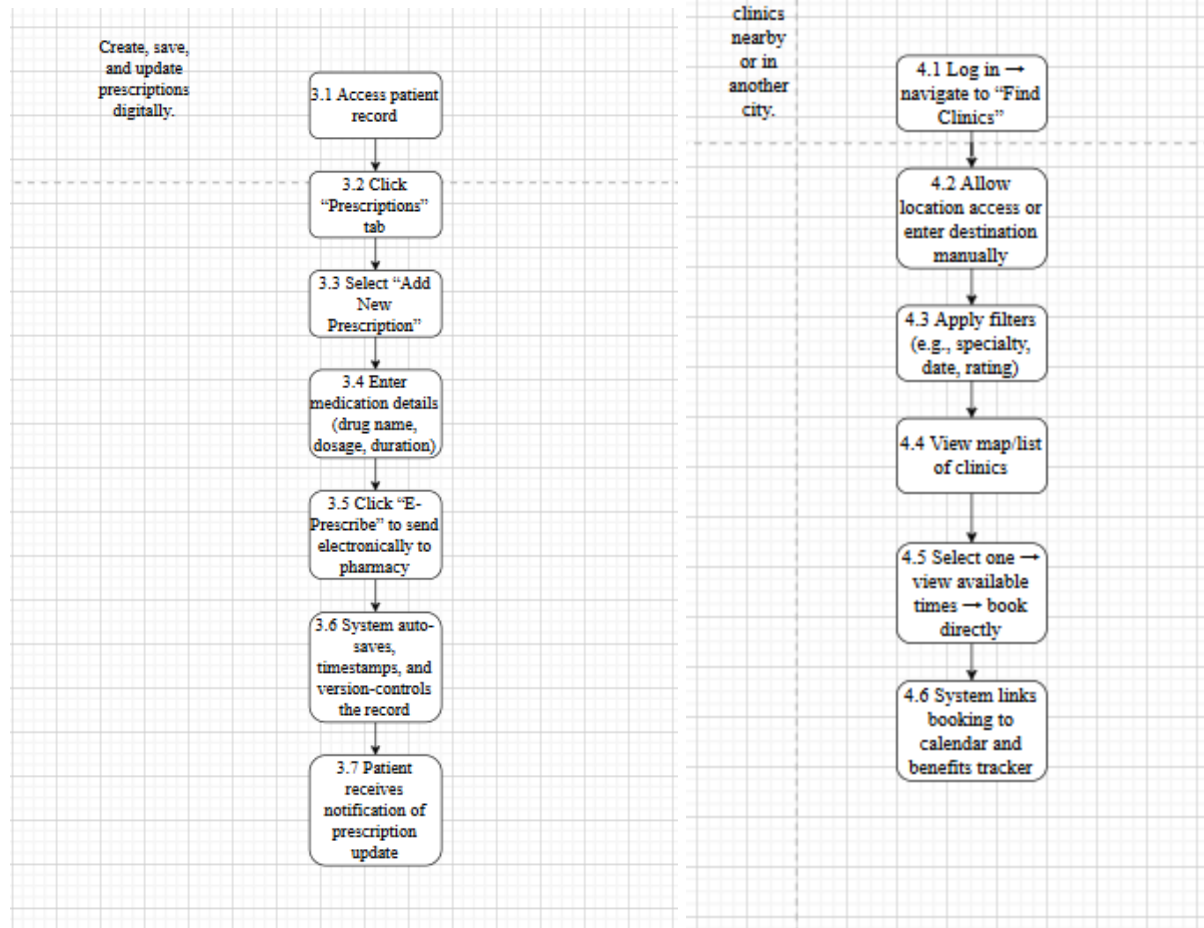


Access patient medical
information efficiently.



Securely
modify
personal or
professional
information.





2. Technique(s) used to gather user requirements with an explanation of why the technique was used (interview, survey, observation, etc.).

We adopted a user-centred design approach to ensure our healthcare application is designed around actual user needs. To understand the tasks, goals, and concerns of users with the existing healthcare systems, we decided to use surveys and semi-structured interviews to collect requirements. Our goal was to create a consistent set of specifications that would guide both the development and evaluation of the prototypes.

Early on, we focused on users and tasks, closely examining behavioural, cognitive, anthropomorphic, and attitudinal characteristics. For example, through our surveys, we explored patients' cognitive inclinations by asking how they interpret health information, how they recall appointment details, and whether they prefer visual accessibility features like flexible text or dark mode. We also observed behavioural characteristics by asking our survey participants how they currently book their appointments or access their medical records.

Additionally, we used empirical measurement by observing, documenting, and analyzing users' responses and actions to scenarios, guides, simulations, and prototypes. For example, interviews with physicians and medical personnel were conducted to determine how fragmented platforms impact their efficiency.

Surveys were provided to users who would serve as patients for our application. These took approximately 4-5 minutes to complete, and the estimated time was mentioned at the start, so participants do not feel overwhelmed by the time. We also promised anonymity to encourage honest answers about personal usage of health applications. We chose surveys because they are relatively easy and quick to distribute, and responses are usually received quickly. Data can also be collected in the Google Forms database with charts and graphs, making it simpler for us to analyze the information, hence reducing the time for data analysis.

We chose to interview healthcare professionals like doctors, nurses, and clinic staff. Interviews provide more accurate and high-quality responses, allowing us to explore more complex issues. We followed a semi-structured approach when conducting the interviews, guided by a script but allowing for flexibility when necessary to explore interesting issues in more depth. This way, interviewees could elaborate and give additional insight, providing a good balance between richness and replicability.

Together, these techniques make sure that our design choices are based on user needs, facilitating the development of a cohesive, reliable, and intuitive medical application that tackles the fragmentation in digital health.

3. Type of users selected, number of users selected and why (describe any limitations).

For our user-centred design process, we selected two primary user groups: healthcare professionals and patients. The interviews focused on doctors, medical staff, and clinical staff, as they are the primary users in charge of maintaining health information, diagnosing conditions, and coordinating care. To obtain a range of perspectives from various medical professionals, we planned to do 10 interviews, trying to establish the validity and diversity of our results. The expert insights gathered from these interviews provided a deeper understanding of workflow inefficiencies, data fragmentation, and inter-system communication barriers, all of which have a direct impact on patient outcomes.

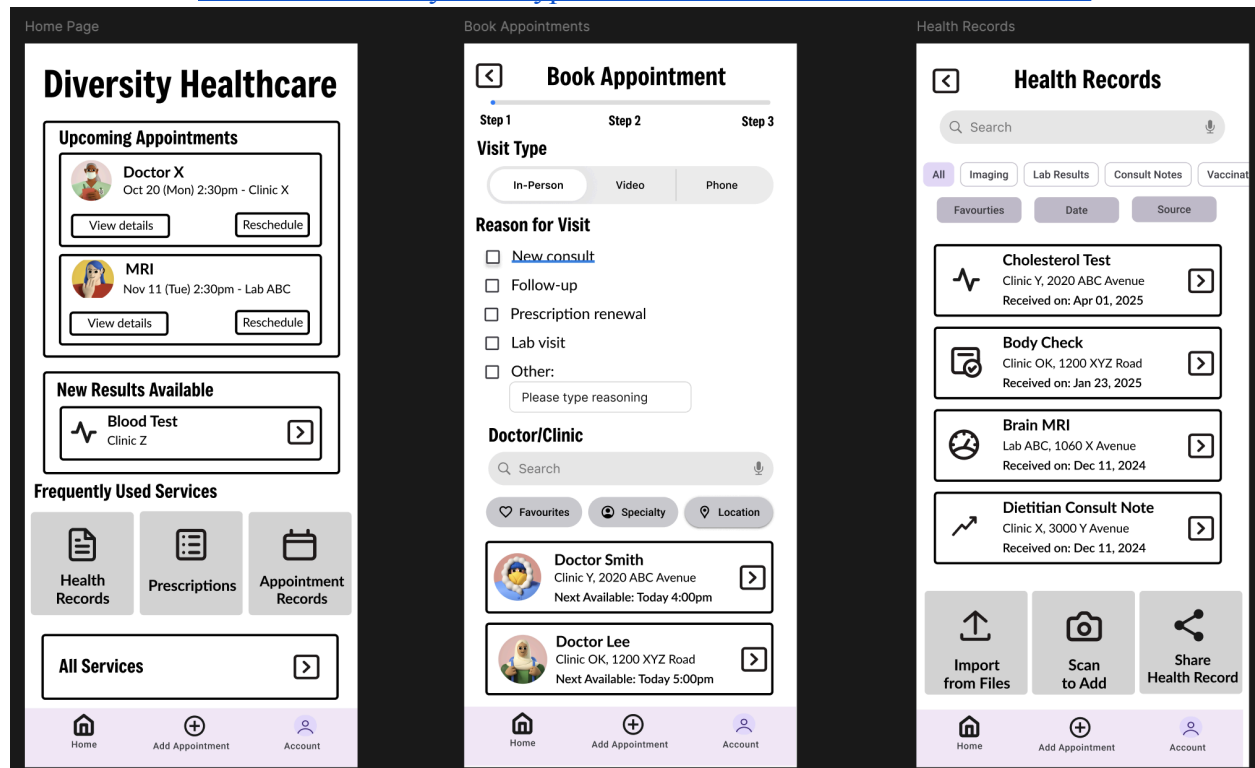
The survey aimed to receive 30 responses from existing users of health applications and potential patients who would be using our app. Patients are the app's end-users, who use it daily to schedule appointments, view records, and manage their health information. This is why we needed a large sample size for the survey to uncover common problems as well as understand user preferences, such as login difficulties, appointment delays, and accessibility requirements, like larger font or dark mode.

This mixed sampling technique helped the design process by ensuring that the requirements were influenced by both the views of medical professionals and patients, giving us an understanding of how we could cater to all types of users. The sample size and demographic diversity were the primary limitations. The participants might not accurately reflect all healthcare settings or patient groups, especially those from low-income or rural areas, where the accessibility needs may be different from those living in dense urban communities.

4. Artifacts used to conduct the user requirements gathering (i.e., sample interview sheet, sample questionnaire, guidelines for assessing observations, low-fidelity prototypes, etc.).

A. Low-fidelity prototype

- <https://www.figma.com/design/JzSfXohCs5Vx8RQYOLftH7/CSCC10-Project-Phase-2-Low-Fidelity-Prototype?node-id=0-1&t=7KTSTJ8bNerT1lf2-1>



B. Sample Interview Sheet

- https://docs.google.com/document/d/1afmvtsybcTOOHH6YpMQfOecCZ_g7LgQk2OmFkNFs5JAI/edit?usp=sharing

Data Privacy, Ethics, and Trust

1. What are your concerns about storing or accessing patient data digitally?
2. How much control do you think patients and doctors should each have over who sees their data?
3. What level of security or verification do you expect before accessing patient files?
4. Would you prefer biometric login, multi-factor authentication, or simple password access?

Usability and Experience

1. When using mobile apps, what design aspects matter most to you — simplicity, speed, dark mode, large text, or others?
2. How much time would you ideally like to spend updating or reviewing patient data in the app per patient?
3. Would you prefer structured templates (e.g., dropdowns and fields) or free-text input when entering patient details?
4. Would offline access to patient data be helpful (for example, in areas with poor network)?
5. Would you like the app to remind you of follow-up visits or pending reports?

Future-Oriented / Open-Ended Questions

1. How do you think digital tools like this could improve healthcare outcomes or your own efficiency?
2. What would make you *not* want to use such an app (e.g., time consumption, privacy issues, poor UX)?
3. How do you think patients will benefit from an app that connects directly with doctors?
4. Would you like to be involved in testing early prototypes and giving feedback as a medical student user?

Background and Context

1. Can you tell me a bit about your role and daily routine as a medical student or intern?
2. How do you currently manage patient data or case notes during your postings or clinical practice?
3. What kind of tools, apps, or hospital systems (if any) do you currently use for managing patient information?
4. How often do you interact with digital health platforms (like lab portals, telemedicine apps, or hospital EMRs)?

Pain Points with Current Systems

1. What challenges do you face while accessing or updating patient information in your current setup?
2. Have you ever experienced confusion or data loss due to fragmented platforms (like separate apps for labs, prescriptions, and history)?
3. How easy or difficult is it to share patient information securely between doctors or departments?
4. When you need a patient's past data (lab results, prescriptions, history), how quickly can you usually find it?
5. Do you find any existing apps unnecessarily complicated or slow during patient care or learning?

Expectations and Feature Preferences

1. If you could design a perfect medical data app for your use, what would it include?
2. What's the most important feature for you — quick access to records, appointment tracking, communication with patients, or something else?
3. Would you find it useful to have AI-assisted summaries (e.g., automatic generation of patient history or treatment suggestions)?
4. Would integration with lab results, prescriptions, or imaging data be valuable for you?
5. How important are data visualization features (like patient trends, vitals over time, or medication adherence graphs)?
6. Would you want an educational dashboard (like clinical reference, treatment protocols, or drug databases) integrated into the same app?

Low-Fidelity Prototype:

1. Appointment Booking Flow:
Does the booking flow (Visit Type → Reason → Provider → Time) match how you expect patients to book follow-ups? What feels missing (e.g. triage, slot duration, buffers, wait-list)?
2. Bookings: Would you prefer a visible time-slot grid or the current list format
Type: time-slot grid/list (filters: Soonest, Virtual-only, Day/Time)
3. Health Records: Pick two primary actions at the bottom of Records.
Type: Choose 2 of Scan to PDF, Request from clinic, Import from Files, Share
4. Homepage: Which Frequently Used Services (Quick Action) would you tap most?
Type: Choose 1 of Prescriptions / Health Records / Appointment Records

C. Sample Questionnaire/Survey

- <https://docs.google.com/forms/d/e/1FAIpQLScRRI0IrrRIzrX0rw7Auwlmv86wQvIE7uavVRj1jQomLNiuVg/viewform?usp=sharing&oid=116518508106687107645>

Healthcare Mobile App – Patient Booking & Records Questionnaire

This anonymous 4–5 minute questionnaire has 8 MCQs, and helps us design a mobile app for booking medical appointments and viewing/sharing health records for CSCC10.

We don't collect names or IDs. Answers will be used only for class research to prioritize features (e.g. calendar syncing, record sharing, accessibility, etc).

Thank you :D

ashleyhwong@gmail.com [Switch account](#)

Not shared

* Indicates required question

How do you book appointments now? *

☐ Clinic portal/website

☐ Phone call

☐ Walk-in

☐ Other: _____

What are the biggest struggles in your current booking process? *

☐ Finding an available time

☐ Multiple logins/portals

☐ Too many steps on phone

☐ Long wait times

☐ Language barriers

☐ Other: _____

Which booking aids would help most in a healthcare app? *

☐ Filter by day/time

☐ Show soonest available

☐ Show next-available by doctor

☐ Show nearby locations

☐ Other: _____

Would you like booked visits added to your phone calendar automatically? *

☐ Yes, always

☐ Ask me each time

☐ No

[Next](#)

[Clear form](#)

Healthcare Mobile App – Patient Booking & Records Questionnaire

ashleyhwong@gmail.com [Switch account](#)

Not shared

* Indicates required question

Health Records

How do you currently receive health records and new results? *

☐ Email Attachments

☐ Paper Copies from visit

☐ Messaging Apps/SMS

☐ Hospital/Clinic/Lab Online Portal

☐ Other: _____

What do you most often do with records? *

☐ View on phone

☐ Download to Files

☐ Share with my doctor

☐ Share with caregiver/family

☐ Add personal notes

☐ None

[Back](#) [Next](#) [Clear form](#)

Healthcare Mobile App – Patient Booking & Records Questionnaire

ashleyhwong@gmail.com [Switch account](#)

Not shared

* Indicates required question

Preferences & Accessibility

What is your preferred sign-in method on secured mobile apps? *

☐ FaceID/TouchID to unlock; password only on first login

☐ 6-digit app PIN to unlock; password only on first login

☐ Password everytime

What accessibility options should a healthcare app have? *

☐ Adjustable text size (Larger texts)

☐ High contrast/dark mode

☐ Screen reader support (auditory support)

☐ Adjustable button size (Larger buttons)

☐ None

☐ Other: _____

[Back](#) [Submit](#) [Clear form](#)

5. Summary of results - explain how your results inform your design decisions as you move to the next phase of the project (i.e., creation of high-fidelity prototypes)

Summary for Survey:

Overview of the findings:

The Healthcare App Survey was filled out by 29 participants in total. The answers showed distinct patterns in the way people now use healthcare systems, schedule appointments, and access medical records. In general, participants complained about restricted accessibility options, unreliable record access, and inconsistent booking systems.

In addition to the survey, eleven interviews with a wide range of participants from different professional backgrounds and ages were conducted. Qualitative information about everyday healthcare management methods, problems, and expectations for a unified healthcare app was gathered from these interviews.

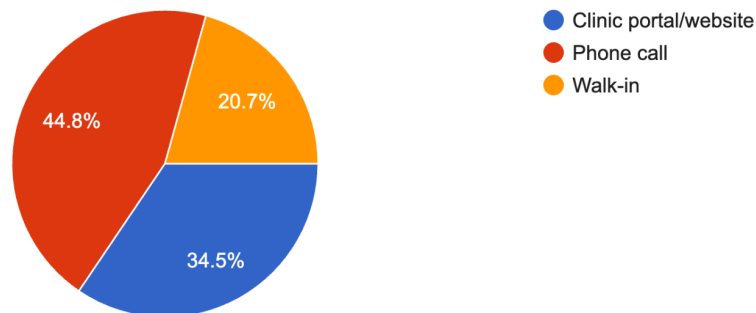
Key findings from the survey:

Appointment Booking

- 46.2% of respondents currently book appointments by phone call, 34.6% through online portals, and 19.2% with walk-ins.

How do you book appointments now?

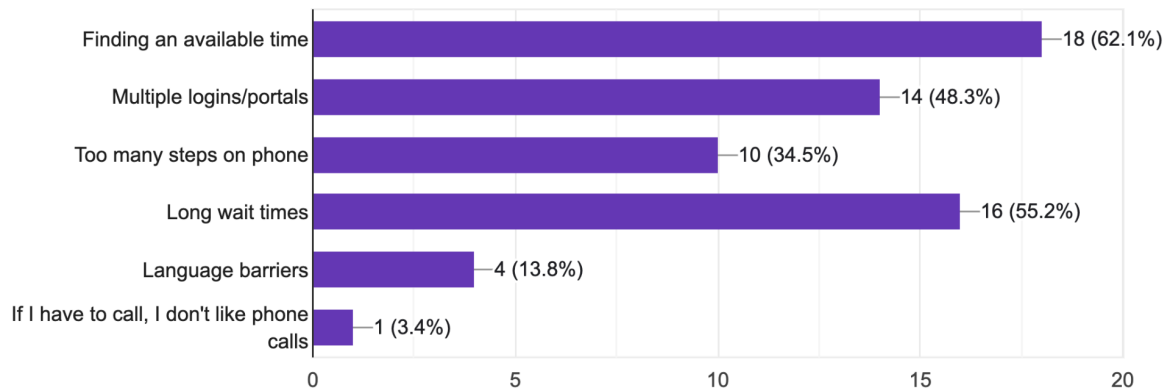
29 responses



- The main struggles included finding available times (57.7%), long wait times (53.8%), and having to manage multiple logins or portals (46.2%).

What are the biggest struggles in your current booking process?

29 responses



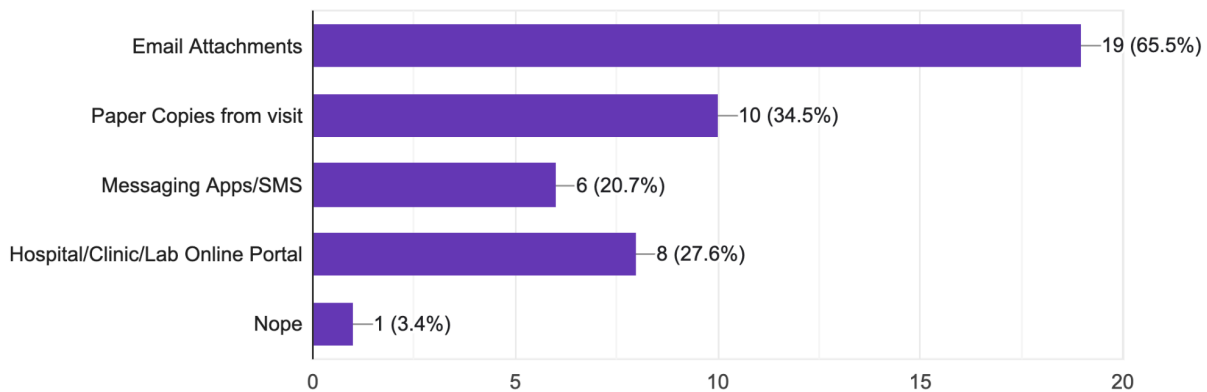
- Design: The app's scheduling interface will give priority to an easy-to-use calendar design that shows the closest clinics, the earliest available times, and the next available by doctor. Instead of navigating several platforms, users would be able to book appointments immediately through one unified platform.

Health Record Access

- Most respondents currently receive their health records by email (65.5%) or paper copies (34.5%), while only 27.9% access them through hospital portals.

How do you currently receive health records and new results?

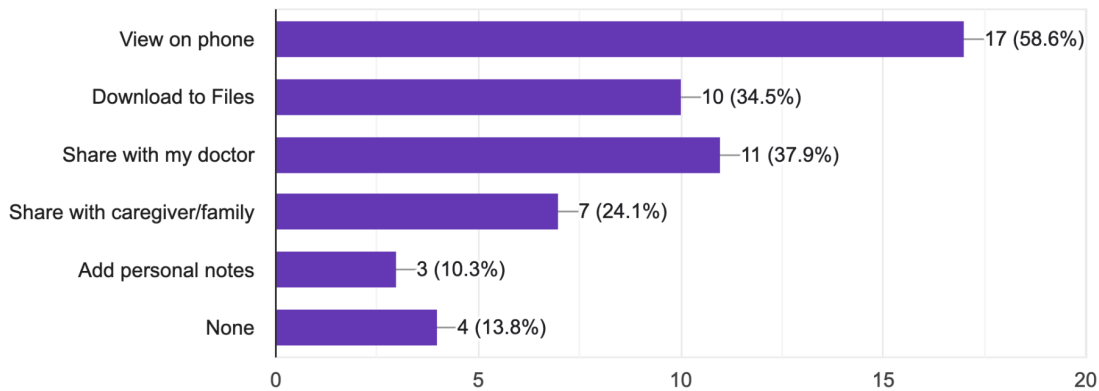
29 responses



- When asked what they do with their records, 58.6% said they view them on their phone, and 37.9% share them with their doctor.

What do you most often do with records?

29 responses



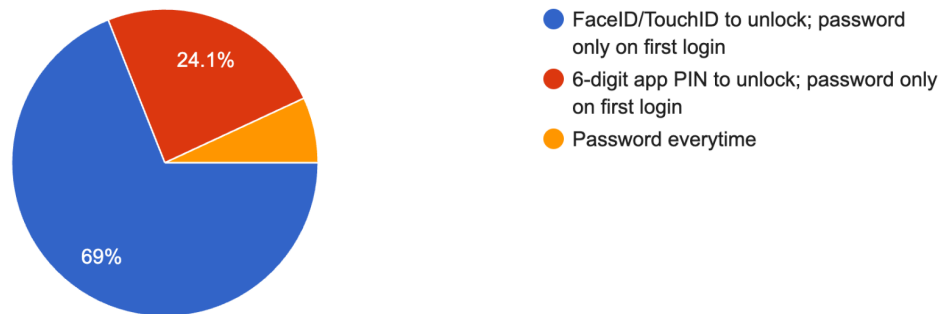
- Design: Prescriptions, visit summaries, and lab results will all be collected in one place through the single app. Users will be able to safely download, view, and share records with staff or medical professionals. To assist users in interpreting information like test results and medication history, the records page will also provide visual summaries.

Accessibility & Security

- The preferred sign-in method was Face ID/Touch ID (69%), followed by PIN and password entry.

What is your preferred sign-in method on secured mobile apps?

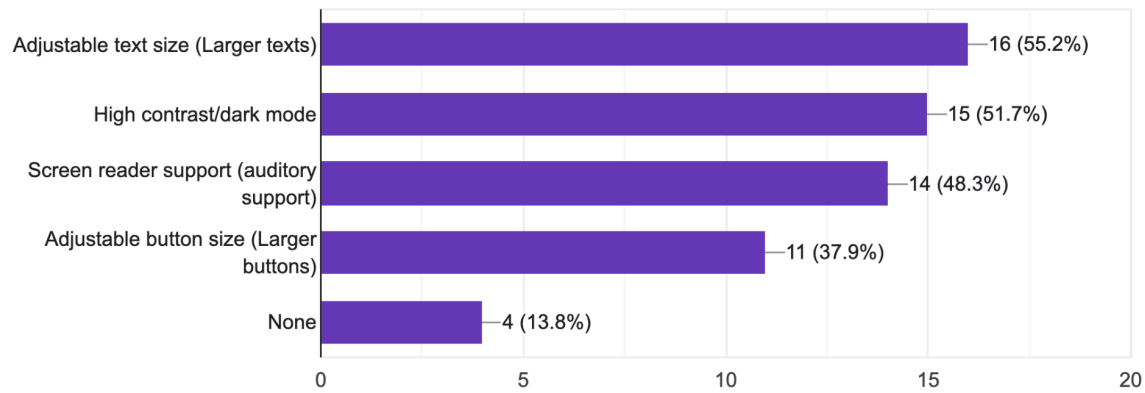
29 responses



- Top accessibility features requested were adjustable text size (55.2%), high-contrast mode (51.7%), and screen reader support (48.3%).

What accessibility options should a healthcare app have?

29 responses



- Design: For quick and safe access, the app will use biometric authentication and adhere to accessibility guidelines. Usability for a wide range of users will be ensured by features like screen reader support, larger buttons, customizable font, and dark mode.

Summary for Interviews: Medical Students (Doctor User Group)

Overview:

To understand the needs, expectations, and challenges faced by medical professionals in training, we conducted semi-structured interviews with 11 MBBS students (6 from 3rd year and 5 from 2nd year). These participants were selected because they represent future end-users of our application — doctors who regularly manage patient data, coordinate care, and navigate fragmented health systems.

The interviews were conducted over individual online calls lasting 15–25 minutes. We explored their current practices, pain points, and expectations from a unified healthcare application that manages both doctor and patient data.

Background and Current Practices

Most participants reported that they currently rely on manual or semi-digital methods to record and retrieve patient data.

- 8 out of 11 mentioned maintaining handwritten case sheets during clinical postings.
- 5 use Google Sheets or WhatsApp to store and share basic notes or case details.
- None had access to a full-fledged hospital EMR (Electronic Medical Record) system.

“During postings, we just note everything in our logbooks or random notebooks. Later, it’s hard to find or recall patient details when we need them for discussions.”
— Miru, 3rd-year

Several students expressed frustration at the lack of centralized systems — they need to access lab data, prescriptions, and appointment info from multiple sources.

“Each lab has a different portal, and hospitals have different record-keeping styles. If we could get all of that in one app, it would make life easier.” — Aiysha, 2nd-year

Pain Points Identified

From the discussions, the following **key challenges** emerged:

Challenge	% of students mentioning it	Example insight
Fragmented patient information (multiple systems, poor data flow)	10/11	“We can’t see a patient’s full history easily.”
Time wasted on paperwork or retrieving data	9/11	“Documentation takes so long; we just skip it sometimes.”
Poor accessibility or usability of existing tools	8/11	“Apps are too cluttered or slow on mobile.”
Lack of secure and easy data sharing among peers/doctors	7/11	“We use screenshots to share lab reports — that’s not safe.”
Limited offline access	5/11	“Wi-Fi is patchy in wards. An offline mode would be great.”

Students consistently described frustration over how data fragmentation reduced their learning efficiency and affected patient care continuity.

Expectations and Desired Features

Students were highly engaged when asked to imagine the “ideal healthcare app.” Their suggestions were grouped into five main feature clusters:

A. Integrated Patient Record System (11/11)

Every participant emphasized the need for a single platform combining:

- Basic patient demographics
- Medical history and prescriptions
- Lab results, radiology, and past visit summaries

“Just give us one space where we can see all the data clearly — history, labs, reports, and diagnosis — that’s all we need.” — Hemanth, 3rd-year

B. Visual Analytics and Summaries (8/11)

Students expressed interest in charts or summaries of patient data over time (vitals, lab trends, medication adherence).

“Graphs for BP or glucose over time would help us spot patterns faster.” — Anand, 2nd-year

They also felt AI-generated summaries could reduce manual effort.

“An AI summary of the last 3 visits or treatment response — that’d be amazing, especially during ward rounds.” — Adithya, 3rd-year

C. Doctor–Patient Communication (7/11)

Most suggested including:

- Secure chat or consultation window
- Automated reminders for appointments or test follow-ups

“It’s better if patients can send updates or questions without calling every time.” — Divya, 2nd-year

D. Privacy and Access Control (9/11)

Participants were cautious about data sharing and stressed the need for:

- Role-based access (doctor vs patient view)
- Multi-factor authentication for sensitive records

“Only authorized doctors should see sensitive details. We must respect patient privacy.” — Naveen, 3rd-year

E. Usability and Design Preferences (10/11)

Most participants valued:

- Simple, uncluttered interface
- Dark mode and flexible font sizes
- Quick search for patient files

“Keep it clean — no fancy animations, just fast and clear.” — Harshini, 2nd-year

F. Low-Fidelity Prototype

- Q1 Appointment Booking Flow: 10/11 confirmed the booking flow (Visit Type → Reason → Provider → Time) was clear.
- Q2 Bookings: 9/11 preferred a visible time-slot grid over a list, and 8/11 specifically wanted inline filters (Soonest, Virtual-only, Day/Time). Qualitatively, 7/11 also recommended reason-based triage and default slot durations/buffers to reduce wrong-type bookings.
- Q3 Health Records: 9/11 selected Scan to PDF and Request from clinic as the two primary persistent actions (over Import/Share).
- Q4 Homepage: 6/11 chose Health Records to be most likely in Frequently User Service (Quick Action), followed by 4/11 for Appointment Records and 1/11 for Prescriptions.

Data Privacy and Trust

Students displayed a high awareness of confidentiality norms due to their coursework on medical ethics.

- 8/11 felt patients should be able to decide who can access their records.

- Several requested audit trails (showing who viewed what and when).

“Transparency builds trust. Even if I’m the doctor, I should justify why I’m accessing a file.” — Rohit, 3rd-year

Learning & Clinical Use

Interestingly, 6 students highlighted potential use cases beyond patient care, such as:

- Using anonymized patient records for academic case studies
- AI-based learning dashboards for differential diagnosis practice

“If I could pull up sample cases for study, it would help in viva or case discussions.” — Sanya, 3rd-year

Key Insights and Common Themes

<i>Theme</i>	<i>Summary Insight</i>
Centralization	All participants want a single unified system integrating all medical data sources.
Ease of use	Doctors prefer speed, minimal clicks, and a clean design over flashy visuals.
Security & Ethics	High importance is given to confidentiality, auditability, and role-based access.
Smart assistance	AI-based summarization and reminders are highly valued for efficiency.
Educational potential	Students see this app as both a professional and a learning tool.

Limitations

- Participants were all students from urban or tertiary hospitals, possibly more tech-savvy than rural practitioners.

- Feedback reflects early-career perspectives, not those of senior consultants.
- Internet reliability was mentioned as an issue, but deeper infrastructure constraints weren't explored.

Conclusion

The interviews revealed a clear desire for an intuitive, secure, and unified healthcare platform that minimizes data fragmentation and supports both care delivery and clinical learning.

In addition to making their patient management responsibilities easier, the interviewees wanted a tool that would improve efficiency, transparency, and collaboration.

Design Choices for High-fidelity Prototype:

Their collective feedback will directly guide our app's feature prioritization — particularly around:

- Unified patient profiles
- AI-generated summaries
- Role-based data privacy
- Simple, mobile-first UI

By anchoring our design choices on these findings, we ensure the system remains practical, ethical, and genuinely user-centred.

Based on the survey and interviews:

- The high-fidelity prototype will include:
 - An appointment schedule that clearly displays the soonest available times, the next available by doctor, and nearby clinic options.
 - A centralized health record dashboard combining lab results, prescriptions, and visit summaries in one accessible view.
 - Automatic calendar syncing for booked visits to help users stay organized.
 - Biometric login (Face ID/Touch ID) for quick, secure access to personal data.

- Design according to accessibility guidelines, including adjustable text size, dark mode, and clear visuals for easy readability.
- Simplified navigation flow with reducing unnecessary steps for common tasks like viewing results or booking appointments.
- More specifically with the feedback given from the low-fidelity prototype:
 - Homepage (Quick Actions order): Prioritize Health Records, then Appointment Records, then Prescriptions, reflecting tap intent.
 - Booking Appointments Page: Keep the 3-step flow; implement a time-slot grid with inline filters (Soonest, Virtual-only, Day/Time), and apply reason-based triage with default durations/buffers.
 - Health Records Page: Replace bottom buttons with a single “Add record” floating action button opening a sheet (Scan • Request • Import). Move “Share” to the record detail view.
- Next-phase priorities:
 - Ensure that users can complete their primary tasks (view results, book appointments, check prescriptions) within two to three taps.
 - Improve clarity of information presentation, especially in medical records and results pages.
 - Maintain consistent visual and interaction patterns throughout the app to improve usability.
 - Conduct usability testing with users of different ages and digital experience levels to ensure ease of navigation and accessibility performance.